Requirement Document: N8N Hands-On Beginner Course

Topic 6: Data Transformation

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 6 - Data Transformation
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
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2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to reformat, clean, and map raw data into a structured output schema using core n8n nodes. By the end of this topic, the learner will be able to perform data cleanup tasks such as renaming keys, trimming whitespaces, converting text casing, and normalizing inputs into standardized properties.

2.2 Learning Scope

In Scope:

- Renaming raw incoming data keys to match a standard destination schema.
- Removing leading and trailing whitespaces from string values using JavaScript's `.trim()` method.
- Normalizing text casing by converting text to uppercase (using `.toUpperCase()`) and lowercase (using `.toLowerCase()`).
- Combining field mapping and text manipulation in n8n expressions.

- Filtering out unnecessary fields by not passing them through the final node.

Out of Scope:

- Complex data validation rules (e.g., regex-based email syntax checks, phone number format normalization).
- Working with nested arrays or object flattening techniques.
- Utilizing third-party scripting libraries for string manipulations.

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 5.
- Familiarity with basic JavaScript string manipulation functions.
- An active n8n instance.

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Dirty Mock Data	The workflow shall start with a node configured to output three mock fields representing dirty, unformatted customer data: `Raw_Name` (with leading/trailing spaces and irregular casing), `contact_email` (with spaces and uppercase characters), and `phone_number` (with spaces).	**High**
REQ-002	Schema Mapping	The workflow shall map the raw incoming keys to standard keys in the output: • `Raw_Name` shall map to `cleanName` • `contact_email` shall map to `cleanEmail` • `phone_number` shall map to `cleanPhone`.	**High**
REQ-003	Whitespace Trimming	The workflow shall strip any leading and trailing whitespaces from all three text strings.	**High**
REQ-004	Case Standardization	The workflow shall convert the output name to uppercase and the output email to lowercase.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-005	Final Payload	The final output shall be a single flat JSON object containing only the mapped keys (`cleanName`, `cleanEmail`, and `cleanPhone`), with all unmapped original fields excluded.	**High**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Chained Javascript Methods	The n8n expression parser must support chaining JavaScript string methods (e.g., `.trim().toUpperCase()`) on data fields.
TR-002	Output Selectivity	The workflow must use an Edit Fields (Set) node configured to not pass all raw inputs through, ensuring only the newly defined clean keys are emitted.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn data hygiene and payload mapping, which are critical skills when preparing payloads for database updates or API requests.
Instructor / Reviewer	Evaluation	Needs to verify that the student renamed fields, removed formatting issues, and correctly applied text casing changes dynamically.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The raw input values are guaranteed to be populated (non-null) strings, avoiding null-pointer expression failures.
- **Constraint 1 (CON-001):** The cleanup rules are statically defined inside the expression panel and do not dynamically inspect field types.